1. What is Cybersecurity? The CIA Triad

What is cybersecurity?

Cybersecurity is the practice of protecting computers, networks, applications and data from attacks, damage or unauthorized access. It is not only about stopping hackers. It also covers accidents (someone deleting a database by mistake), system failures and insider misuse.

In simple words: **making sure the right people can use the right information at the right time, and nobody else can.**

- Protects personal data (Aadhaar details, bank accounts, medical records)
- Keeps businesses running (a hospital or a bank cannot afford downtime)
- Covers people, process and technology, not just software
- Works in three ways: **prevent** attacks, **detect** them when they happen, **respond and recover** after

The CIA Triad

The CIA triad is the base model for almost everything in security. Every control we put in place is trying to protect at least one of these three things.

Part	Meaning	Real-world example
Confidentiality	Only authorized people can see the data	Your WhatsApp messages are end-to-end encrypted, so even the company cannot read them. If someone leaks your chats, confidentiality is broken.
Integrity	Data stays accurate and is not changed without permission	You transfer Rs. 5,000 through UPI. If an attacker changed the amount to Rs. 50,000 on the way, integrity is broken. Hashes and digital signatures help here.
Availability	Systems and data are accessible when needed	A DDoS attack takes down an online ticket booking site on the day bookings open. The data is safe, but nobody can use it. That is an availability failure.

Quick notes

- The three often pull against each other. Very strict access control improves confidentiality but can hurt availability.
- Different systems care about different parts. A bank cares most about integrity, a news website cares more about availability.
- Common controls: encryption (C), hashing and checksums (I), backups and redundancy (A).